PYTHON PROGRAMMING ASSIGNMENT

Build Your Own Rule-Based Python Chatbot

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TASK	**30-Minute Student Task**

1. Aim

To develop a simple rule-based conversational chatbot in Python. The chatbot continuously interacts with the user, recognizes different messages, provides suitable responses, and allows the user to exit the conversation.

2. Python Program

```
# Rule-Based Python Chatbot
# Name: Shaik Thanvir Ahmed
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print("Welcome to My Python Chatbot!")

name = input("What is your name? ").strip()
print("Hello", name + "! Let's start chatting.")
print("Type bye or exit whenever you want to stop.")

while True:
    message = input("You: ").lower().strip()

    if message == "bye" or message == "exit":
        print("Bot: Goodbye", name + "! Have a nice day.")
        break

    elif "hello" in message or "hi" in message:
        print("Bot: Hello", name + "! How can I help you?")

    elif "python" in message:
        print("Bot: Python is a simple and powerful programming language.")

    elif "how are you" in message:
        print("Bot: I am doing great. Thanks for asking!")

    elif "your name" in message:
        print("Bot: I am a rule-based chatbot made with Python.")

    elif "weather" in message:
        print("Bot: I cannot check live weather, but I hope it is nice today.")

    else:
        print("Bot: Sorry, I do not understand. Please try another message.")
```

3. Working of the Program

Concept	Function
input()	Gets the user's name and messages.
.lower().strip()	Cleans and standardizes the user's message.
while True:	Keeps the chatbot conversation running.

Concept	Function
if / elif / else	Chooses a response based on the message.
in operator	Detects greetings even when extra words are present.
break	Ends the conversation when bye or exit is entered.

4. Sample Output

```
Welcome to My Python Chatbot!
What is your name? Thanvir
Hello Thanvir! Let's start chatting.

You: hello bot
Bot: Hello Thanvir! How can I help you?

You: tell me about python
Bot: Python is a simple and powerful programming language.

You: how are you
Bot: I am doing great. Thanks for asking!

You: random
Bot: Sorry, I do not understand. Please try another message.

You: bye
Bot: Goodbye Thanvir! Have a nice day.
```

5. Requirement Checklist

✓	Asks the user's name before the main conversation.
✓	Uses while True to keep the conversation running.
✓	Uses input() to receive messages.
✓	Uses .lower().strip() to clean input.
✓	Recognizes hello and hi greetings.
✓	Answers multiple questions/topics.
✓	Uses if / elif / else decision making.
✓	Uses break for bye and exit.
✓	Uses the user's name in responses.
✓	Uses else for unknown messages.

6. Conclusion

The chatbot follows the required INPUT → PROCESS → RESPOND → REPEAT → EXIT flow. It demonstrates basic Python programming concepts such as input handling, string methods, loops, conditional statements, the in operator, and break.

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